THE CECH COMPLEX FUNCTOR ON ABELIAN PRESHEAVES IS EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. The same-chapter pointer is located at its \label line inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- The Cech complex functor on abelian presheaves of a fixed family. Same chapter, the displayed definition labelled equation-cech-functor (source line 913, Stacks tag 03AP); the statement refers to it by equation number. Supplies the functor whose exactness is asserted and the shape of its degree p terms as products of evaluation functors.
- Exactness criterion for functors between abelian categories. Homology, Lemma labelled homology-lemma-exact-functor (canonical Stacks tag 010N); named in the counted statement itself. Supplies the criterion that has to be verified; the verification for the Cech complex functor is the counted content.

Source reference. Original equation 03AP: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 913–913. Same-chapter pointer to the displayed definition of the Cech complex functor, named in the counted statement at line 927; the exactness argument is retained verbatim in the counted span.

Source reference. Original lemma 010N: sources/stacks-chow/tags.txt, source lines 1333–1333. Cross-chapter pointer named in the counted statement at line 928 for the criterion of exactness of a functor; the verification of that criterion is retained in the counted span.

Lemma 0.1. *The functor given by Equation (03AP) is an exact functor (see Homology, Lemma 010N).*

Proof. For any object W of $\mathcal{C}$ the functor $\mathcal{F} \mapsto \mathcal{F}(W)$ is an additive exact functor from $PAb(\mathcal{C})$ to Ab. The terms $\check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F})$ of the complex are products of these exact functors and hence exact. Moreover a sequence of complexes is exact if and only if the sequence of terms in a given degree is exact. Hence the lemma follows. □

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